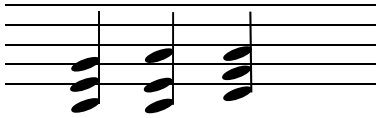
Homework 2 (Due: 29th Oct.)

(1) What are the main advantages and the disadvantages of the recursive method for implementing the STFT? (10 scores)

(2) Calculate the WDF of $\cos(6\pi t + 1))$ (10 scores)

(3) For which of the following functions the WDFs may have the cross term problem? Why? (a) $\delta(-\pi t)$; (b) $\exp(-\pi t^4)$; (c) $\exp(j\pi t^4)$; (d) $\cos(-\pi t^2)$; (e) A music signal  . (15 scores)

(4) Why (a) Cohen's class distribution, (b) the polynomial WDF, and (c) the Gabor-Wigner transform can avoid the cross term problem in some cases? (15 scores)

(5) (a) Sort the computation loading of the following time-frequency analysis methods from large to small (i) rec-STFT, (ii) Gabor-Wigner transform; (iii) windowed WDF; (iv) Cohen's class distribution. (b) What is the constraint for the window $w(\tau)$ to make the output of the windowed WDF a real function? (c) Is the windowed WDF a one-to-one operation? Why? (15 scores)

(6) Write a Matlab or Python program for the rectangular STFT. (35 scores)

(the window is $w(t) = 1$ if $|t| < B$, $w(t) = 0$ otherwise).

y = recSTFT(x, t, f, B)

x: input, *t*: samples on *t*-axis, *f*: samples on *f*-axis, *y*: output

(i) The code (*.m or *.py) should be handed out by NTUCool

(ii) Choose an input x. Plot the output y.

(iii) Use tic and toc to determine the running time

(iv) Write by the **FFT-based method.**

Example：

```
dt=0.05; df=0.05;
t1=[0:dt:10-dt]; t2=[10:dt:20-dt]; t3=[20:dt:30];
t=[0:dt:30];
f=[-5:df:5];
x=[cos(2*pi*t1),cos(6*pi*t2),cos(4*pi*t3)];
tic
y=recSTFT(x, t, f , B);
toc
```

(Extra): Answer the questions according to your student ID number.

(ended with 0, 1, 2, 4, 5, 6, 7, 9)